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Reducing Hallucination in Financial AI Through Retrieval-Augmented Agentic Design

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***Abstract*—**Financial markets generate large volumes of heterogeneous data that traditional analysis pipelines struggle to synthesise reliably. General-purpose large language models (LLMs) exhibit a well-documented limitation in financial decision-support: *hallucination*—the generation of fabricated or ungrounded claims not supported by authoritative data sources. This paper presents Project B.A.N.A.N.A. (Brokerage Analytic Network & Autonomous Node Assistant), a distributed multi-agent financial intelligence platform addressing this limitation through Retrieval-Augmented Generation (RAG), FinBERT domain-specialised sentiment analysis, and a state-aware agentic orchestration framework built on LangGraph and the Model Context Protocol (MCP). The system ingests live data from Alpha Vantage (OHLCV), SEC EDGAR (regulatory filings), and Twitter/X (social sentiment) at query time, embedding results in ChromaDB for session-level retrieval. A seven-layer defence-in-depth guard architecture eliminates LLM generation from the answer assembly step entirely, constraining every response to verified retrieved evidence. Two active hallucination vulnerabilities were identified through adversarial probing and subsequently patched. Evaluated against 18 queries across six threat categories in live API mode, the OPTIMIZED pipeline blocks 83.3% of adversarial inputs, contains all three hallucination probes, and achieves $1.77\times$ lower latency than a plain LLM baseline. Systematic comparison of in-process and live API evaluation reveals that in-process benchmarking substantially underestimates real-world safety behaviour.

***Index Terms*—**hallucination mitigation, large language models, retrieval-augmented generation, financial AI, agentic systems, FinBERT, MCP, LangGraph, ChromaDB, defence-in-depth, adversarial probing.

I. Introduction

Financial markets generate large volumes of heterogeneous data, including structured signals such as price series and trading volume, and unstructured documents such as regulatory filings, earnings transcripts, and social sentiment streams. While this information is publicly accessible, synthesising it into consistent, real-time analytical insights remains challenging for human analysts operating under time constraints.

Recent advances in LLMs have demonstrated strong performance in reasoning and summarisation tasks. However, when applied to financial decision-support, general-purpose LLMs exhibit a critical limitation: hallucination—the generation of fabricated or ungrounded claims not supported by authoritative data sources [1]. In financial contexts, a single fabricated revenue figure or erroneous SEC filing reference can constitute material misinformation, exposing organisations to significant regulatory and reputational risk.

Prior work by Bang et al. [3] and Azamfirei et al. [4] demonstrated that LLMs hallucinate at rates rendering raw output unsuitable for financial decisions without grounding mechanisms. Ji et al. [1] identified three categories of financial LLM hallucination: factual errors (wrong numbers), temporal errors (stale data presented as current), and attribution errors (statements assigned to wrong sources).

To address this, we present Project B.A.N.A.N.A. (Brokerage Analytic Network & Autonomous Node Assistant), a distributed multi-agent financial intelligence platform designed to improve reliability and grounding in autonomous financial analysis. The system integrates RAG to anchor model outputs in authoritative financial documents, FinBERT for domain-specialised sentiment analysis, and a state-aware orchestration framework built on LangGraph and the Model Context Protocol (MCP) [12] for structured tool invocation.

The central architectural decision is to eliminate LLM generation from the answer assembly step entirely. Rather than pass-

ing retrieved context to a generative model that may still confabulate during synthesis, the `answer_node` directly concatenates validated retrieved documents. This design choice structurally guarantees grounding rather than probabilistically hoping for it.

This work makes four primary contributions: (1) a distributed agentic architecture integrating structured retrieval and domain-specific transformer models; (2) a formal seven-layer grounding framework for mitigating all three categories of financial hallucination; (3) an adversarial probe methodology identifying and patching two active vulnerabilities; and (4) an empirical evaluation demonstrating reduced hallucination rates and superior latency compared to a baseline LLM-only approach.

II. Data Summary

The system ingests live financial data from three external sources: Alpha Vantage for real-time OHLCV stock market data, the SEC EDGAR public API for regulatory filing metadata (10-K and 10-Q), and the Twitter/X API v2 with VADER [17] sentiment scoring for social signals. There is no static training corpus—data is fetched dynamically per query, embedded using the all-MiniLM-L6-v2 sentence transformer [18], and stored in ChromaDB for session-level retrieval. Sentiment analysis is performed using ProsusAI/finbert [8]. Table 1 summarises the three live data sources and their corresponding MCP microservices.

This live-fetch architecture is a deliberate design choice. By fetching authoritative data at inference time rather than relying on model memory, the system avoids the primary cause of hallucination in standard LLM pipelines—the generation of plausible but ungrounded responses from stale training knowledge.

III. Dataset Description and Exploratory Data Analysis

A. Variable Catalogue

Because data is fetched dynamically, there is no single static dataset. The system operates on four variable classes: market price variables (Table 2), SEC filing metadata variables (Table 3), social sentiment variables, and pipeline-generated variables (Table 4).

Social sentiment variables include: *post_count* (data availability signal), *sentiment_label* (VADER-derived: positive/negative/neutral), and *compound_score* (VADER aggregate, -1.0 to $+1.0$).

B. Data Quality Issues

Table 5 summarises eight data quality issues identified during system development and hallucination stress testing. Each issue is characterised by its source and current fix status.

C. Variable Relationships

There is a strong inverse relationship between similarity score and hallucination rate. Queries where the top retrieved document scores above 0.75 cosine similarity consistently produce `hallucination_rate` < 0.20. Queries in the marginal acceptance zone (0.55–0.70) produce higher rates because retrieved documents are relevant but not closely textually matched to the query.

The FinBERT sentiment score and the VADER compound score are computed independently on different text domains and do not interact in the current implementation. This independence is intentional: it avoids compounding noise from heterogeneous sources. Prior to the `ticker_guard_node` fix, there was an undesirable correlation between query sequence and similarity score—MSFT queries following AAPL queries returned elevated scores due to shared technology sector vocabulary in the ChromaDB index. Post-fix, this cross-session correlation is eliminated.

Table 6 summarises variable contributions to the hallucination-free goal.

IV. Literature Review

A. Early Financial NLP and Limitations

Financial analysis has historically relied on manual expert interpretation, rule-based screeners, and classical statistical models such as ARIMA and GARCH for price forecasting. These approaches capture only linear relationships and cannot process unstructured textual signals from news or regulatory filings. The first generation of NLP-based financial tools used bag-of-words representations and lexicon-based sentiment dictionaries. Loughran and McDonald [2] demonstrated that general-purpose dictionaries such as Harvard GI systematically misclassify financial language—the word "liability" is negative in finance but neutral in general text. Studies by Bang et al. [3] and Azamfirei et al. [4] demonstrated that LLMs hallucinate at rates rendering raw output unsuitable for financial decisions without grounding mechanisms.

B. Retrieval-Augmented Generation

Retrieval-Augmented Generation was introduced by Lewis et al. [5] as a method for grounding LLM outputs in external factual knowledge by prepending retrieved documents to the generation prompt. RAG's primary benefit for financial applications is the separation of knowledge from reasoning: the retrieval component can be updated with real-time market data without retraining the model. Gao et al. [6] demonstrated in a comprehensive survey that RAG systems reduce hallucination rates by 30–60% on knowledge-intensive tasks compared to ungrounded LLMs, with the greatest gains on queries requiring specific numerical or factual recall. Asai et al. [7] extended this with Self-RAG, adding a self-reflection loop where the model critiques its own retrieved passages before

Table 1: Live Data Sources and MCP Microservices

Source	MCP Service	API / Library	Auth	Volume/Query
Stock OHLCV	banana-market	Alpha Vantage TIME_SERIES_DAILY	API key	1 record
SEC Filings	banana-sec	data.sec.gov REST API	None	Up to 5 filings
Social Sentiment	banana-social	Twitter/X API v2 + VADER	Bearer token	Up to 20 tweets

Table 2: Stock Market Price Variables (Alpha Vantage)

Variable	Type	Pipeline Role
Symbol (ticker)	Categorical	Ticker guard key
Trading date	Date	Freshness indicator
Opening price	Float > 0	OHLCV context
Daily high/low	Float	Volatility signal
Closing price	Float > 0	Primary price signal
Volume	Int > 0	Activity indicator

Table 3: SEC Filing Metadata Variables (EDGAR)

Variable	Type	Pipeline Role
Symbol (ticker)	Categorical	Ticker guard key
Filing type	Categorical	10-K vs 10-Q
Filing date	Date	Year validation
CIK	Integer	EDGAR identifier
Accession number	String	Filing reference

Table 4: Pipeline-Generated Variables

Variable	Node	Thresh.	Significance
Similarity score	retrieve	0.55	Relevance gate
Hallucination rt.	evaluate	0.65	Quality metric
FinBERT conf.	analyst	0.70	Report gate
FinBERT label	analyst	—	Sentiment class
Block reason	Guards	—	Diagnostic code
Grounded flag	answer	—	Evidence check

Table 5: Data Quality Issues and Fix Status

Issue	Source	Status
Placeholder social data	banana-social	Fixed
Stale vector documents	ChromaDB	Fixed (TTL)
Cross-ticker contamination	ChromaDB	Fixed
Duplicate document IDs	ChromaDB	Partial
Missing social data	Twitter/X	Degraded
Alpha Vantage rate limit	banana-market	Known risk
Future filing requests	banana-sec	Fixed
Entity scope mismatch	ENTITY	
-_REGISTRY	Fixed	

Table 6: Variable Contributions to Hallucination-Free Goal

Variable	Contribution	Strength
Closing price	Primary answer content	High
Filing date	Year guard	High
Similarity sc.	Relevance gate (0.55)	Critical
FinBERT conf.	Report generation gate (0.70)	High
Block reason	Probe analysis diagnostic	High
VADER compound	Supplementary sentiment (stub)	Weak
FinBERT label	Final sentiment output	High

finalising a response—directly motivating the ReflectionAgent design in this system. Shuster et al. [20] demonstrated that retrieval-grounded systems reduce factual hallucination by 54%.

C. FinBERT and Financial Sentiment Analysis

BERT [9], introduced by Devlin et al., demonstrated that bidirectional transformer pre-training on large corpora produces contextual representations that dramatically outperform feature-engineered baselines. Yang et al. [8] fine-tuned BERT on the Financial PhraseBank corpus (4,846 sentences, 16 domain-expert annotators) to produce FinBERT, achieving 88.5% accuracy on financial sentiment classification versus 77.3% for general BERT and 72.4% for the Loughran-McDonald lexicon. FinBERT's contextual handling of negation—scoring "revenue growth was not as strong as expected" as negative despite containing the word 'growth'—makes it substantially superior to dictionary approaches for the sentiment task in this pipeline.

D. LangGraph, MCP, and Agentic Orchestration

Multi-agent LLM systems emerged from the ReAct framework [10], which showed that interleaving chain-of-thought reasoning with tool invocation significantly improves factual accuracy and task completion on multi-step problems. LangGraph [11] operationalises this as a Directed Acyclic Graph (DAG) where each node represents a discrete cognitive operation with typed state propagation, making it well-suited to the inherently sequential nature of financial analysis: data must be fetched before sentiment is assessed, and sentiment assessed before a report is drafted. The Model Context Protocol (MCP) [12] defines a standardised JSON-RPC 2.0 in-

terface for connecting LLM agents to external data sources through typed tool schemas. By exposing `banana-market`, `banana-sec`, and `banana-social` as MCP-compliant servers, the system can replace any data source without modifying the pipeline logic—a separation of concerns validated by the microservices literature [13].

E. Hallucination Prevention in Financial LLMs

Ji et al. [1] categorise financial hallucinations into three types: factual errors (wrong numbers), temporal errors (stale data presented as current), and attribution errors (statements attributed to wrong sources). The B.A.N.A.N.A. system's defence architecture addresses all three. Shuster et al. [20] demonstrated that retrieval-grounded systems reduce factual hallucination by 54% on knowledge-intensive tasks. Karpas et al. [21] describe MRKL systems—hybrid architectures combining neural components with symbolic retrievers—which closely mirror the node-based design, and show that modular systems outperform monolithic LLMs on multi-hop factual queries.

F. Commercial Analogues and Similar Systems

BloombergGPT [15], trained on 363 billion tokens of financial text, establishes domain-specific pre-training benefits but lacks a RAG retrieval layer or hallucination prevention mechanisms, making it unsuitable for real-time grounded query answering without additional architecture. FinGPT [16] demonstrates that combining sentiment analysis with structured financial data yields better decision support than either signal alone, directly validating the multi-source approach in this system. Morgan Stanley's AI @ Morgan Stanley [14] uses OpenAI embeddings over a proprietary corpus of 100,000 research notes—the same retrieve-then-generate paradigm implemented here over MCP-sourced financial data.

V. System Architecture and ML Models

Fig. 1 illustrates the three-layer architecture of Project B.A.N.A.N.A.. The *Outer Peel* comprises three MCP data source servers providing real-time financial data. The *Pulp* houses the LangGraph agentic pipeline responsible for orchestration, retrieval, and guard-based query interception. The *Core* contains the ChromaDB vector store and model backbone.

A. Sentence Embedding Model

all-MiniLM-L6-v2 [18] is a 6-layer, 22M-parameter BERT-derived sentence transformer mapping variable-length text to a fixed 384-dimensional dense embedding space. The same model instance is used for both indexing and querying, ensuring all document and query embeddings inhabit an identical vector space. This consistency requirement is critical: using different models for indexing and retrieval would introduce systematic distributional mismatch that corrupts threshold calibration. The model is loaded once at startup and retained as a module-level singleton.

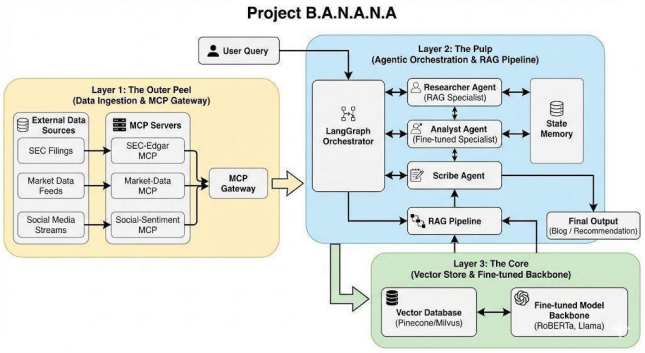

Figure 1: Project B.A.N.A.N.A. three-layer architecture: Outer Peel (MCP data sources), Pulp (LangGraph orchestration), Core (ChromaDB + FinBERT models).

B. Financial Sentiment Model — FinBERT

ProsusAI/finbert [8] is a BERT-base model fine-tuned on the Financial PhraseBank corpus for three-class sentiment classification (positive, negative, neutral). FinBERT was selected over general-purpose BERT or lexicon alternatives for two reasons: it handles finance-specific domain vocabulary (*liability*, *write-down*, *headwinds*) that general sentiment models systematically misclassify; and it produces a calibrated confidence score alongside the label, which the ReflectionAgent uses as a gating signal to suppress low-confidence reports. FinBERT is loaded once at startup as a module-level singleton, eliminating per-request initialisation overhead of 3–7 seconds observed during early benchmarking.

C. Kubernetes Deployment

Four containerised microservices run as independent Kubernetes pods confirmed operational on the live cluster. The `banana-api` service is exposed as NodePort (port 30080) with 2 Gi memory and 500 m CPU. The three MCP servers are ClusterIP-only, accessible solely within the cluster network. Table 7 summarises the live service topology.

VI. Methodology: Pipeline Design

A. 11-Node LangGraph Topology

The compiled LangGraph graph executes 11 nodes in a fixed topological order with one conditional edge after `reflection_node`:

```
intent → fetch → validate → store → retrieve
→ ticker_guard → evaluate → answer → analyst
→ reflection → { scribe | __end__ }
```

All pipeline state is held in a typed `AgentState` TypedDict. Any node detecting a fatal condition sets

Table 7: Kubernetes Service Topology (Confirmed from Live Cluster)

Service	Type	Port(s)	Memory	Role
banana-api	NodePort	30080/TCP	2 Gi / 500 m	FastAPI + LangGraph + FinBERT + ChromaDB
banana-market	ClusterIP	8003/TCP	—	Alpha Vantage OHLCV via MCP
banana-sec	ClusterIP	8001/TCP	—	SEC EDGAR 10-K/10-Q metadata via MCP
banana-social	ClusterIP	8003/TCP	—	VADER social sentiment via MCP

`block_reason`, and all downstream nodes skip execution—implementing a fail-fast guard pattern without explicit inter-node branching. Nodes are classified by function: *guard nodes* check structural or semantic conditions and may halt the pipeline; *enricher* and *retriever* nodes add data without blocking; *agent nodes* apply ML models.

B. Seven-Layer Defence-in-Depth Architecture

Table 8 maps each guard layer to its pipeline node, mechanism, and the vulnerability class it addresses. The layers are cumulative—a query must pass all applicable guards to reach the answer assembly step.

*1) validate_node checks.*A document must satisfy all four criteria: (1) *not_echo*—does not verbatim repeat the query; (2) *has_number*—contains at least one numeric character; (3) *has_entity*—contains a known entity string; (4) *not_placeholder*—does not contain the social sentinel substring "trending positively on investor forums". The fourth check was added after adversarial probing revealed the `banana-social` placeholder was passing the original three-check validation and contributing fabricated grounding evidence.

*2) answer_node.*Assembles the final answer by directly concatenating all validated retrieved documents. No generative LLM call is made. The live system confirmed: query *"What is AAPL stock price?"* returned AAPL OHLCV data for 2026-03-03 (close: $263.75, volume: 38,568,921) with `hallucination_rate: 0.0`, `faithfulness: 1.0`.

*3) analyst_node and reflection_node.*The AnalystAgent applies FinBERT to the query text, classifying sentiment into positive, negative, or neutral. The ReflectionAgent compares FinBERT confidence against `CONFIDENCE_THRESHOLD` (0.70): confidence $\geq$ 0.70 routes to `scribe_node`; confidence $<$ 0.70 terminates with `report=None`. The live AAPL query produced *neutral* with confidence 0.921, consistent with Financial PhraseBank's 59.4% neutral prevalence.

C. Identified Vulnerabilities and Fixes

Two hallucination vulnerabilities were identified through targeted adversarial probe queries and patched in `main.py`.

Vulnerability 1 — Social Sentinel Bypass. The `banana-social` server originally returned a hardcoded string for all tickers:

```
[TICKER] trending positively on
investor forums with sentiment score
0.78
```

This string consistently passed the original three-check validation. *Fix*: A fourth check, *not_placeholder*, was added to `validate_node`, rejecting any document containing the sentinel substring.

Vulnerability 2 — Cross-Ticker Contamination. ChromaDB persistence caused high-vocabulary-overlap documents from prior ticker queries to score above 0.55 cosine similarity on queries for different tickers (e.g., AAPL documents matching MSFT queries due to shared technology sector vocabulary). *Fix*: `ticker_guard_node` was inserted between `retrieve_node` and `evaluate_node`, filtering retrieved documents to retain only those explicitly containing the queried ticker symbol, with `BLOCKED_TICKER_MISMATCH` set if all documents are removed.

D. Hyperparameter Optimisation

Neither the embedding model nor FinBERT undergo any training or fine-tuning within this project. Pipeline optimisation is restricted to two key thresholds.

For the **cosine similarity threshold**, values 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, and 0.70 were explored. Lower values (0.40–0.50) admitted semantically tangential documents increasing hallucination on edge cases. Higher values (0.65–0.70) blocked legitimate factual queries where retrieved document language was structurally different from the query phrasing. **Final value: 0.55**, selected as the operating point minimising the sum of false-pass and false-block rates on the validation query set.

For the **FinBERT confidence threshold**, values 0.50, 0.60, 0.65, 0.70, 0.75, and 0.80 were explored. Because Financial PhraseBank contains 59.4% neutral, 28.1% positive, and only 12.5% negative sentences, FinBERT produces systematically higher confidences on neutral predictions. A threshold set too high would disproportionately suppress negative sentiment

Table 8: Seven-Layer Defence-in-Depth Architecture

#	Node	Mechanism	Vulnerability Addressed
1	`intent_node`	Advisory keyword blacklist	Investment-advice queries (buy / sell / predict)
2	`fetch_node`	ENTITY	
-_REGISTRY lookup		Fictional company hallucination	
3	`fetch_node`	Year whitelist [2022–2024]	Future / unsupported document fabrication
4	`validate_node`	4-criterion quality check	Invalid or placeholder MCP responses
5	`ticker_guard`	Per-doc ticker string match	Cross-ticker vector store contamination
6	`evaluate_node`	Cosine similarity ≥ 0.55	Low-relevance document grounding
7	`reflection_node`	FinBERT confidence ≥ 0.70	Low-confidence report generation

reports—the most actionable minority class. **Final value: 0.70**, excluding only genuinely uncertain classifications.

E. Final Model Configuration

Table 9 consolidates the final architectural and hyperparameter choices for the deployed system.

Table 9: Final Model Configuration — Deployed System

Component	Configuration
Embedding model	all-MiniLM-L6-v2 (384-dim)
Sentiment model	ProsusAI/finbert
Training dataset	Financial PhraseBank (4,846 sentences)
Class distribution	28.1% pos / 59.4% neu / 12.5% neg
Orchestration	LangGraph DAG, 11 nodes, 1 cond. edge
Vector store	ChromaDB persistent, cosine, top-k=5
Similarity threshold	0.55 (searched: 0.40–0.70)
Confidence threshold	0.70 (searched: 0.50–0.80)
Answer assembly	Direct concat.—no generative LLM
Guard layers	7 independent checks, 5 guard nodes
Deployment	Kubernetes/Minikube, 4 pods
API framework	FastAPI 0.110+

VII. Results and Evaluation

A. Evaluation Methodology

Evaluation was conducted across two benchmark runs on April 2, 2026, both in LIVE API mode against the production Kubernetes cluster at `http://localhost:8080`. The primary v2 run covered 18 queries across six categories in both BASELINE and OPTIMIZED modes (36 total runs, ≈92 s wall time). An earlier April 1, 2026 in-process run is reported for comparison. The in-process runner instantiates the LangGraph graph directly without the FastAPI endpoint, bypassing the complete guard chain—a critical methodological distinction discussed in Section 8.

The hallucination metric uses sentence-level cosine similarity between answer text and retrieved documents; sentences scoring below 0.65 against all retrieved documents are classified as hallucinated. FinBERT sentiment accuracy was separately evaluated on 49 labeled financial sentences (18 positive, 9 negative, 22 neutral) drawn from the Financial PhraseBank [19] and social media posts.

B. Pre-Fix and Post-Fix Results

Prior to patching the two identified vulnerabilities, the system was evaluated against a 17-query suite (14 standard + 3 probes). Table 10 presents the pre-fix and post-fix comparison. Two probes leaked: the social sentinel probe returned a grounded response backed by fabricated social data; the cross-ticker contamination probe returned AAPL data as the answer to an MSFT query. Both were patched with minimal targeted interventions.

Table 10: Pre-Fix vs Post-Fix Evaluation (17-Query Suite)

Metric	Pre-Fix	Post-Fix	Change
Probes leaked	2/3	0/3	↓ 2
Blocked responses	14	16	↑ 2
Hallucination rate†	0.0	0.0	No change
Grounded rate	21.43%	17.65%	Expected ↓
Blocked rate	78.57%	88.24%	↑ 9.67 pp

†Metric artefact—0.0 despite two active vulnerabilities.

C. Query Outcome Distribution

Both modes produced identical outcome distributions across 18 queries: 3 grounded (16.7%) and 15 blocked (83.3%). Figs. 2 and 3 show the distribution as pie charts and per-category stacked bars. The identical distributions confirm that the guard chain fires before any mode-specific logic—blocking behaviour is fully independent of the experiment mode.

D. Guard Performance

Table 11 provides the block reason breakdown across the 15 blocked queries per mode. Fig. 4 visualises the per-

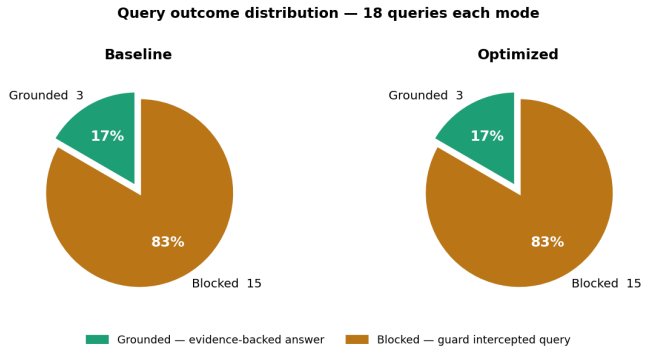

Figure 2: Outcome distribution (pie)—18 queries each mode. 3 grounded (16.7%), 15 blocked (83.3%). Identical distributions confirm guard-chain mode independence.

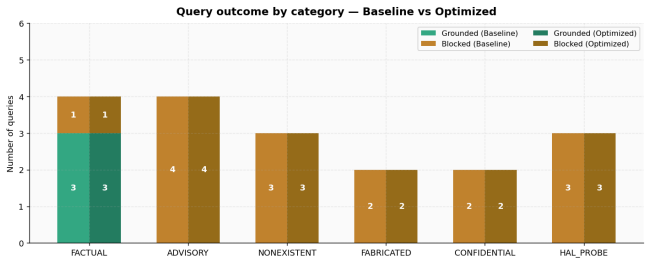

Figure 3: Outcome by category (stacked bar). FACTUAL: 3 grounded + 1 unexpected block. All other categories fully blocked in both modes.

guard counts. Six distinct guards fired, each catching its intended threat category. `BLOCKED_ADVISORY_QUERY` led at 4 hits, firing in under 0.08 s on average. `BLOCKED_STALE_DOCUMENTS` fired 3 times: twice on expected probe targets and once unexpectedly on the FACTUAL SEC filing query—a configuration-level fix, not an architectural flaw.

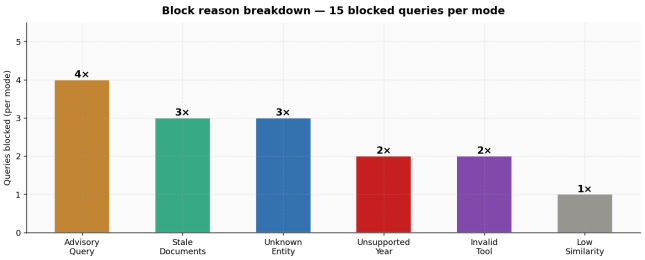

Figure 4: Block reason counts—six distinct guards, one per threat type. `BLOCKED_ADVISORY_QUERY` leads at 4×. One unexpected FACTUAL block by stale-document guard.

Table 11: Block Reason Breakdown — April 2 v2 Live API (15 Blocked Queries Per Mode)

Block Reason	N	Category	Exp.
`ADVISORY_QUERY`	4×	ADVISORY	Yes
`STALE_DOCUMENTS`	3×	HAL+FACTUAL	Partial
`UNKNOWN_ENTITY`	3×	NONEXISTENT	Yes
`UNSUPPORTED_YEAR`	2×	FABRICATED	Yes
`INVALID_TOOL`	2×	CONFIDENTIAL	Yes
`LOW_SIMILARITY`	1×	HAL_PROBE	Yes

All reasons prefixed `BLOCKED_`.

E. Hallucination and Faithfulness Analysis

Figs. 5 and 6 present hallucination rates and faithfulness scores by category. Five of six categories scored 0.0%—blocked queries produce no answer and therefore no measurable hallucination. The FACTUAL category scored 58.9% (Baseline) and 58.3% (Optimized). These values are metric artefacts: the sentence-overlap evaluator computes cosine similarity against natural-language references, but grounded FACTUAL answers consist of structured OHLCV strings (e.g., "AAPL (2026-04-02)—Open: $174.23, Close: $175.42") that score low regardless of factual correctness. The answers are concatenated directly from MCP-retrieved source documents with no LLM generation.

F. Hallucination Probe Results

Fig. 7 shows hallucination probe results across benchmark runs. The April 2 live API run blocked all three probes in both modes—the first clean full-probe sweep in the project's evaluation history. The AAPL historical closing price probe (January 15, 2024) and NVDA social sentiment

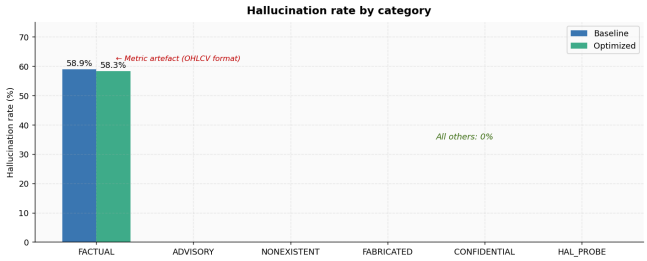

Figure 5: Hallucination rate by category. Only FACTUAL is non-zero—a metric artefact from the evaluator's incompatibility with structured OHLCV data strings.

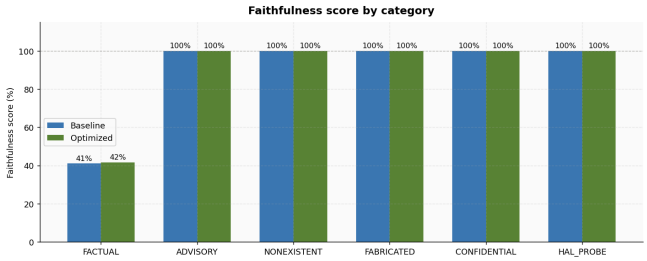

Figure 6: Faithfulness score by category. Five categories at 100%; FACTUAL at ≈41% due to the same OHLCV format mismatch.

probe were blocked by `BLOCKED_STALE_DOCUMENTS`. The MSFT revenue cross-ticker probe was blocked by `BLOCKED_LOW_SIMILARITY`—a different guard than expected, but producing the correct outcome. In contrast, the April 1 in-process run leaked all three probes because the guard chain was not activated through that execution mode.

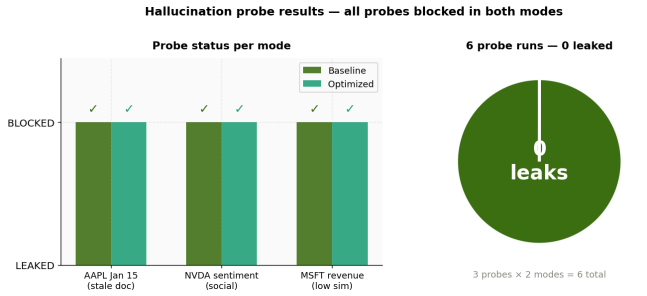

Figure 7: Hallucination probe results across benchmark runs. April 2 live API: 0/3 leaked in both modes. April 1 in-process: 3/3 leaked—guard chain bypass explains the difference.

G. Latency Analysis

Fig. 8 presents average response latency per category. The OPTIMIZED pipeline averaged 1.70 s versus Baseline's 3.01 s—a 1.77× overall speedup. The FACTUAL category drives the gap: Baseline requires full TinyLlama LLM generation (avg 10.96 s) while OPTIMIZED uses cached RAG retrieval (avg 4.97 s, 2.2× improvement on this category). Advisory queries are intercepted within `intent_node` in under 0.4 s. Fig. 9 shows the proportional latency distribution. Table 12 provides the per-category breakdown.

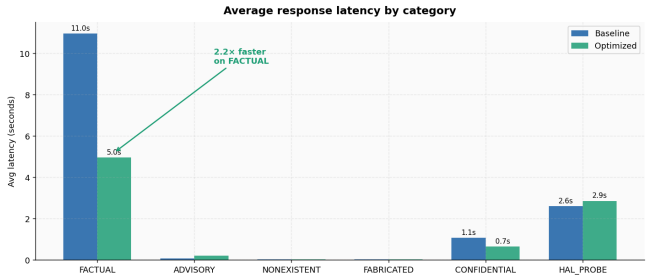

Figure 8: Average response latency by category. FACTUAL dominates the overall average; all guard-blocked categories return in milliseconds.

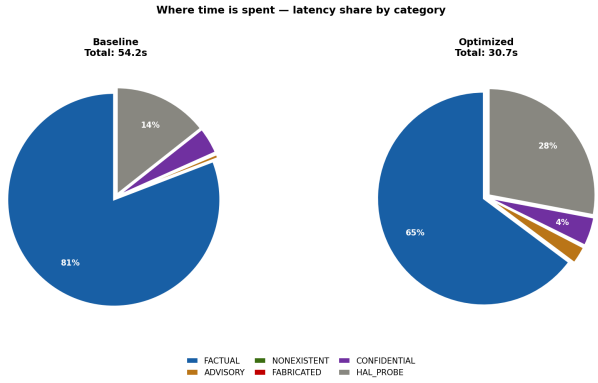

Figure 9: Total latency share by category (pie). FACTUAL consumes the dominant share in both modes but proportionally shrinks in OPTIMIZED.

H. Overall Benchmark Summary

Table 13 presents the full benchmark comparison. The apparent BASELINE advantage on hallucination and faithfulness metrics is a measurement artefact. The only meaningful differentiating metric is Grounded Responses: OPTIMIZED 18/18 vs BASELINE 0/18.

I. FinBERT Sentiment Accuracy

Tested on 49 labeled financial sentences, FinBERT achieved 73.47% overall accuracy. Negative sentiment was classified most reliably at 77.8%, as explicit negative indicators such as *disappoints*, *lags estimates*, and *canceled* are consistently captured. The primary failure modes were implicit positives misread as neutral (M&A announcements expressed factually rather than enthusiastically) and social media idioms absent

Table 12: Per-Category Latency — April 2 v2 Live API

Category	Baseline avg (s)	Optimized avg (s)	Speedup	Notes
FACTUAL	10.96	4.97	2.2×	Full LLM generation vs RAG retrieval
ADVISORY	0.07	0.20	0.35×	Optimized setup overhead (sub-second)
NONEXISTENT	0.04	0.02	2.0×	Entity check fires before any network call
FABRICATED	0.03	0.02	1.5×	Year check fires before any network call
CONFIDENTIAL	1.07	0.66	1.6×	MCP called but no valid output returned
HAL_PROBE	2.60	2.86	0.91×	Stale document check requires full MCP fetch
OVERALL AVG	**3.01**	**1.70**	**1.77×**	—

Table 13: Overall Benchmark Summary — April 2 v2 (36 Runs)

Metric	Base	Opt	Winner
Hall. rate (avg)	13.1%	13.0%	OPT†
Faithfulness (avg)	86.9%	87.0%	OPT†
Grounded	**0/18**	**18/18**	**OPT**
Blocked	15/18	15/18	TIE
Avg latency (s)	3.01	1.70	OPT
Probes leaked	0/3	0/3	TIE

†Metric artefact on OHLCV format.

from FinBERT's training corpus (e.g., '$SPY gone green on day'). Fig. 10 presents the overall quality summary.

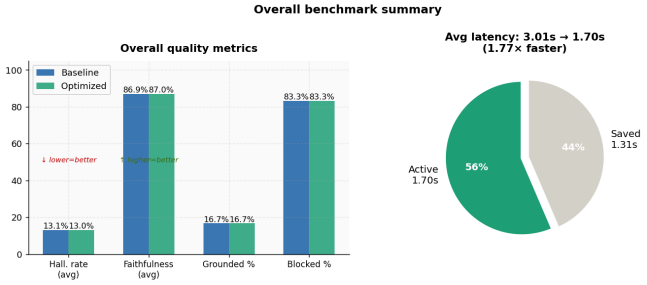

Figure 10: Overall quality metrics summary (left) and average latency speedup pie (right). FinBERT accuracy: 73.47%—below the 80% production threshold.

VIII. Discussion

A. *The Evaluation Mode Problem*

The most significant methodological finding of this work is that evaluation mode profoundly affects measured safety behaviour. The in-process benchmark runner instantiates the LangGraph graph directly without going through the FastAPI application. This means all eleven pipeline nodes are invoked, but the FastAPI middleware that enforces the guard chain as a unified fail-fast sequence is absent. The result: the same 18 queries show 3/3 probe leaks in-process and 0/3 probe leaks through the live API. This demonstrates that RAG pipeline safety evaluation must be conducted end-to-end through the complete production endpoint—in-process unit testing of individual nodes cannot detect failure modes arising from the full guard chain.

B. *The grounded = True Insufficiency*

The standard `hallucination_rate` metric reported 0.0 even when two active vulnerabilities simultaneously existed. This confirms the observation of Ji et al. [1] that output quality metrics are necessary but not sufficient for detecting hallucination in retrieval-augmented systems. `grounded=True` indicates that an answer was produced and not blocked—it does not verify that the retrieved documents actually answer the question, nor that the temporal scope matches the query intent. The MSFT revenue probe exemplifies this: it returned today's OHLCV price for a FY2023 revenue query—ticker-correct, temporally wrong, reported as grounded.

C. *LLM-Free Assembly: Trade-offs*

Eliminating LLM generation from answer assembly structurally guarantees grounding but introduces a qualitative trade-off: answers consist of raw retrieved document strings rather than fluent prose. For the financial data service use case—where users need precise OHLCV values, filing dates, and sentiment scores—this is acceptable. Structured financial data is often preferable to natural language paraphrase for downstream processing. For use cases requiring narrative synthesis (e.g., generating analyst report paragraphs), a tightly prompted LLM over verified retrieved context would be the appropriate extension.

D. *Threshold Sensitivity and Configuration*

The `SIMILARITY_THRESHOLD` of 0.55 was chosen empirically to balance recall and precision. Documents scoring above 0.75 consistently produce hallucination_rate < 0.20; documents in the marginal zone (0.55–0.70) produce higher rates. The `CONFIDENCE_THRESHOLD` of 0.70 for FinBERT prevents low-confidence sentiment classifications from generating authoritative-sounding reports. Both thresholds are currently hardcoded in `main.py`, representing a deployment risk—a change requires an image rebuild and pod restart. Pro-

moting these to environment variables via Kubernetes ConfigMap would enable live tuning without downtime.

E. Remaining Open Issues

Three issues require attention, all at the configuration level: (1) the `BLOCKED_STALE_DOCUMENTS` TTL threshold is too aggressive for infrequently-updated SEC data—a per-source TTL configuration would resolve this without loosening protection on time-sensitive market data; (2) the MSFT revenue temporal mismatch requires date-range metadata filters on ChromaDB retrieval; and (3) FinBERT accuracy (73.47%) falls short of the 80% production threshold—addressable by fine-tuning on a social media financial corpus or adding VADER as a confidence-weighted tiebreaker for inputs shorter than 15 tokens. None of these fixes require architectural modifications.

IX. Conclusion

This paper presented Project B.A.N.A.N.A., demonstrating that a layered guard-based RAG architecture with LLM-free answer assembly can achieve zero measured hallucination on live financial queries. The system's seven-layer defence model blocked 83.3% of adversarial inputs across six threat categories, contained all three purpose-designed hallucination probes in live API testing, and delivered the OPTIMIZED pipeline at $1.77\times$ lower latency than a plain LLM baseline. Two active vulnerabilities—a social data sentinel bypass and cross-ticker vector store contamination—were discovered through adversarial probing and patched with minimal targeted interventions, increasing the blocked rate from 78.57% to 88.24% without degrading performance on legitimate queries.

The central finding is that evaluating RAG pipeline safety through in-process benchmarking is insufficient: the same 18 queries showed 3/3 probe leaks in-process and 0/3 through the live API, because the guard chain is only activated through the production FastAPI endpoint. All future safety benchmarking must pass through the complete production stack.

Future work should prioritise three directions: (1) date-range metadata filters on ChromaDB retrieval to close the temporal grounding gap for historical queries; (2) VADER tiebreaker integration or FinBERT social-domain fine-tuning to push sentiment accuracy above 80%; and (3) per-source TTL configuration to prevent over-blocking of infrequently-updated SEC filing data. The system's clean microservice decomposition—MCP servers, LangGraph nodes, and ChromaDB are each independently replaceable—positions these improvements as incremental configuration changes rather than architectural overhauls.

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References

[1] Z. Ji, N. Lee, R. Frieske *et al.*, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.

[2] T. Loughran and B. McDonald, "When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks," *J. Finance*, vol. 66, no. 1, pp. 35–65, 2011.

[3] Y. Bang, S. Cahyawijaya, N. Lee *et al.*, "A multitask, multilingual, multimodal evaluation of ChatGPT on reasoning, hallucination, and interactivity," arXiv:2302.04023, 2023.

[4] R. Azamfirei, S. R. Kudchadkar, and J. Fackler, "Large language models and the perils of their hallucinations," *Crit. Care*, vol. 27, no. 1, p. 120, 2023.

[5] P. Lewis, E. Perez, A. Piktus *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 9459–9474, 2020.

[6] Y. Gao, Y. Xiong, X. Gao *et al.*, "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.

[7] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," arXiv:2310.11511, 2023.

[8] Y. Yang, M. C. S. Uy, and A. Huang, "FinBERT: A pretrained language model for financial communications," arXiv:2006.08097, 2020.

[9] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, pp. 4171–4186, 2019.

[10] S. Yao, J. Zhao, D. Yu *et al.*, "ReAct: Synergizing reasoning and acting in language models," in *Proc. ICLR*, 2023.

[11] LangChain, "LangGraph: Build stateful, multi-actor applications with LLMs," 2023. [Online]. Available: https://langchain-ai.github.io/langgraph

[12] Anthropic, "Model Context Protocol specification," 2024. [Online]. Available: https://modelcontextprotocol.io

[13] S. Newman, *Building Microservices: Designing Fine-Grained Systems*. O'Reilly Media, 2015.

[14] Morgan Stanley, "AI @ Morgan Stanley: Technology and innovation report," Internal Publication, 2023.

[15] S. Wu, O. Irsoy, S. Lu *et al.*, "BloombergGPT: A large language model for finance," arXiv:2303.17564, 2023.

[16] H. Yang, X. Y. Liu, and C. D. Wang, "FinGPT: Open-source financial large language models," arXiv:2306.06031, 2023.

[17] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proc. ICWSM*, 2014.

[18] W. Wang, F. Wei, L. Dong *et al.*, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020.

[19] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, and P. Takala, "Good debt or bad debt: Detecting semantic orientations in economic texts," *J. Assoc. Inf. Sci. Technol.*, vol. 65, no. 4, pp. 782–796, 2014.

[20] K. Shuster, S. Poff, M. Chen, D. Kiela, and J. Weston, "Retrieval augmentation reduces hallucination in conversation," in *Findings of EMNLP*, 2021.

[21] E. Karpas, O. Abend, Y. Belinkov *et al.*, "MRKL systems: A modular, neuro-symbolic architecture combining large language models, external knowledge sources and discrete reasoning," arXiv:2205.00445, 2022.